

SmartStore ERP — Backend Stack

Recommended Technology Stack for Custom Backend | Prepared for Kartik (Alphenta)

Recommended Stack

Layer	Technology	Why This Choice
API Server	FastAPI (Python)	Fastest Python framework, auto-generates API docs, async support, easy to read
Database	PostgreSQL	Free, reliable, handles GST/accounting math perfectly, scales from 1 store to 1000
Authentication	JWT Tokens	Lightweight, works with HTML frontend, supports role-based access (owner/cashier/staff)
Offline / PWA	Service Worker + IndexedDB	Works without internet, syncs when online — critical for tier-2/3 Indian cities
File Storage	Local Disk or S3	For invoices, PDF reports, data backups
Real-time Updates	WebSocket	Live dashboard KPIs, instant POS updates across devices
Hosting	Any VPS (Rs 500/mo)	DigitalOcean, Hetzner, or even a local mini-server in the shop

Why FastAPI + PostgreSQL (Not Node.js/Express)

- ✓ **Python = easier for AI features** — AI Demand Forecast, Fraud Detection, Chatbot NLP all need Python libraries (pandas, scikit-learn, numpy). With Node.js you'd need a separate Python service anyway.
- ✓ **FastAPI auto-generates Swagger docs** — you can see and test every API endpoint in your browser at /docs without writing any documentation. Perfect for debugging and understanding the backend.
- ✓ **PostgreSQL handles money correctly** — DECIMAL type means no floating-point errors in GST calculations. Rs 280.00 stays Rs 280.00, not Rs 279.9999997.
- ✓ **Same language for everything** — backend API + AI models + data processing + Tally export + PDF generation = all Python. No context switching between languages.
- ✓ **Massive ecosystem** — libraries for everything Indian: GST validation, UPI, Razorpay SDK, WhatsApp Business API, Tally XML — all in Python.
- ✓ **Easy to hire for** — if you ever need a developer, Python developers are 3x more available than Node.js in India.

How It All Connects

Your 18-Module HTML Frontend (Glassmorphism UI, Marathi + English)

↓ fetch() calls (JSON) ↓

FastAPI Backend (Python) Auth | APIs | Business Logic | AI Models

↓ SQL queries ↓

PostgreSQL Database All 18 modules, permanent storage

Every button in your UI that currently shows a toast message will instead call a real API, which reads/writes real data, stored permanently in PostgreSQL.

What Each Layer Handles

Layer	Handles	Examples
FastAPI	Authentication	Login, JWT tokens, role check (owner vs cashier)
FastAPI	CRUD APIs	Create invoice, add product, update stock, delete expired
FastAPI	Business Logic	GST calculation, FIFO costing, udhaar interest, scheme triggers
FastAPI	AI Models	Demand forecast (scikit-learn), fraud scoring, chatbot NLP
FastAPI	Integrations	Razorpay UPI, Gupshup WhatsApp, Tally XML export
FastAPI	PDF Generation	Thermal print invoices, GST reports, bank reconciliation
PostgreSQL	Master Data	Products, suppliers, dealers, staff, HSN codes, barcodes
PostgreSQL	Transactions	Sales, purchases, payments, credit notes, stock movements
PostgreSQL	Ledgers	Accounting entries, GST ledger, udhaar ledger, bank entries
PostgreSQL	Analytics	Daily summaries, monthly reports, trend data for AI
Service Worker	Offline Cache	POS works without internet, syncs when online
Service Worker	Background Sync	Queued transactions push to server when WiFi returns
WebSocket	Live Updates	Dashboard KPIs refresh in real-time, multi-device POS sync

All 19 Gaps Solved By This Stack

#	Gap	How This Stack Solves It
1	No real backend	FastAPI + PostgreSQL = full backend
2	No offline/PWA	Service Worker + IndexedDB
3	No mobile app	PWA manifest = installable on phone
4	No multi-location	Warehouse/location tables in PostgreSQL
5	No role-based access	JWT + role middleware in FastAPI
6	No real payment gateway	Razorpay Python SDK for UPI/Card
7	No PO approval workflow	Maker-checker logic in FastAPI endpoints
8	No delivery/GRN tracking	GRN table + status tracking in DB

#	Gap	How This Stack Solves It
9	No FIFO/weighted avg costing	Python calculation on stock_movement table
10	No customer loyalty/CRM	Customer table + points/tier logic
11	No actual e-Invoice API	NIC portal API integration via Python requests
12	No Tally export	Python XML generator for Tally import format
13	No real bank reconciliation	CSV parser + auto-matching algorithm in Python
14	No print invoice	ReportLab PDF for thermal/A4 print
15	WhatsApp one-way only	Gupshup/Twilio 2-way API in Python
16	AI chatbot hardcoded	Real DB queries behind chatbot responses
17	No supplier rating	Supplier scoring from PO/GRN data
18	No returns/exchange	Credit note + return stock entry APIs
19	Branding decision	Config table — switch name anytime

Cost Summary

Item	Cost	Notes
FastAPI + Python	Free	Open source
PostgreSQL	Free	Open source
VPS Server	Rs 500-800/month	DigitalOcean/Hetzner, 2GB RAM enough
Domain + SSL	Rs 500-1000/year	SSL free via Let's Encrypt
Razorpay	2% per transaction	Standard UPI/Card gateway fees
WhatsApp API	Rs 0.50-1 per message	Gupshup Business API pricing
Claude (backend dev)	Free	I build it in conversation
TOTAL	Rs 600-1,000/month	Vs ERPNext Rs 2,000-5,000/month

SUMMARY

FastAPI (Python) + PostgreSQL + Service Worker

- **One language (Python)** for backend + AI + integrations + exports
- **One database (PostgreSQL)** for all 18 modules — no data splitting
- **Your UI untouched** — just add fetch() calls to real APIs
- **All 19 gaps solved** — from auth to Tally export to offline PWA
- **Rs 600-1,000/month total cost** — affordable for small mall clients
- **You test, Claude builds** — minimal effort on your side

